Week 3: Lab

Data preprocessing, wrangling, and visualization

2025-11-17

Load packages

```
pkgs <- c("tidyverse", "dplyr", "readr", "magrittr", "ggplot2",
          "cowplot", "GGally", "knitr", "data.table",
          "skimr", "flextable", "Metrics", "ggpubr", "devtools")

for (i in seq_along(pkgs)){
  pkg <- pkgs[[i]]
  if(pkg%in%rownames(installed.packages())){
    cat("Available:", pkg, "\n")
  }
  else{
    cat("Not available:",pkg,">> now installing...", "\n")
    install.packages(pkg, repos =  "https://cloud.r-project.org")
    cat("Now available:", pkg, "\n")
  }

  library(pkg, character.only = TRUE)
  cat("Package loaded:", pkg,"\n")
}
```

```
Available: tidyverse

Package loaded: tidyverse
Available: dplyr
Package loaded: dplyr
Available: readr
Package loaded: readr
Available: magrittr
```

```
Package loaded: magrittr
Available: ggplot2
Package loaded: ggplot2
Available: cowplot

Package loaded: cowplot
Available: GGally
Package loaded: GGally
Available: knitr
Package loaded: knitr
Available: data.table

Package loaded: data.table
Available: skimr
Package loaded: skimr
Available: flextable

Package loaded: flextable
Available: Metrics
Package loaded: Metrics
Available: ggpubr

Package loaded: ggpubr
Available: devtools

Package loaded: devtools
```

Load the data

```r
liv <-
  "https://raw.githubusercontent.com/Bleznolap/GENET_5910_2025/main/Data/ClinicalTraits.csv"

## Load the data

liver_df <- read_csv(liv)

## Check the dimension: how many rows and columns

dim(liver_df)
```

```
[1] 361  38
```

1). Check the data

```r
## Check the total NAs
sum(is.na(liver_df))
```

```
[1] 1403
```

```r
## Look at the data descriptive statistics
summary(liver_df)
```

```
      ...1          Mice               Number          Mouse_ID        
 Min.   :  1   Length:361         Min.   :  1.0   Length:361        
 1st Qu.: 91   Class :character   1st Qu.: 92.0   Class :character  
 Median :181   Mode  :character   Median :183.0   Mode  :character  
 Mean   :181                      Mean   :182.5                     
 3rd Qu.:271                      3rd Qu.:273.0                     
 Max.   :361                      Max.   :363.0                     
                                                                    
    Strain               sex             DOB                parents      
 Length:361         Min.   :1.000   Min.   :2001-10-22   Min.   :   103  
 Class :character   1st Qu.:1.000   1st Qu.:2002-01-01   1st Qu.:   111  
 Mode  :character   Median :1.000   Median :2002-02-28   Median :   227  
                    Mean   :1.499   Mean   :2002-02-18   Mean   : 22735  
                    3rd Qu.:2.000   3rd Qu.:2002-03-22   3rd Qu.:   230  
                    Max.   :2.000   Max.   :2002-12-30   Max.   :231227  
                                                         NA's   :39      
  Western_Diet           Sac_Date             weight_g       length_cm    
 Min.   :2001-12-17   Min.   :2002-04-08   Min.   :22.20   Min.   : 8.70  
 1st Qu.:2002-02-26   1st Qu.:2002-06-19   1st Qu.:37.70   1st Qu.:10.10  
 Median :2002-04-22   Median :2002-08-21   Median :43.60   Median :10.40  
 Mean   :2002-04-18   Mean   :2002-08-10   Mean   :42.98   Mean   :10.43  
 3rd Qu.:2002-05-14   3rd Qu.:2002-09-10   3rd Qu.:48.80   3rd Qu.:10.70  
 Max.   :2003-02-21   Max.   :2003-06-16   Max.   :64.40   Max.   :12.00  
 NA's   :3                                 NA's   :4       NA's   :4      
     ab_fat        other_fat       total_fat        comments        
 Min.   :0.150   Min.   :0.150   Min.   : 0.320   Length:361        
 1st Qu.:1.170   1st Qu.:1.900   1st Qu.: 3.280   Class :character  
 Median :1.700   Median :2.310   Median : 4.100   Mode  :character  
 Mean   :1.926   Mean   :2.327   Mean   : 4.242                     
 3rd Qu.:2.520   3rd Qu.:2.700   3rd Qu.: 5.175                     
```

```
 Max.   :6.580   Max.   :5.900   Max.   :10.350
 NA's   :4       NA's   :2       NA's   :6
 100xfat_weight       Trigly        Total_Chol      HDL_Chol
 Min.   : 1.265   Min.   :  0.0   Min.   : 380   Min.   : 10.00
 1st Qu.: 7.284   1st Qu.: 54.0   1st Qu.:1058   1st Qu.: 30.00
 Median : 9.385   Median : 85.0   Median :1292   Median : 41.00
 Mean   : 9.942   Mean   :114.1   Mean   :1301   Mean   : 43.39
 3rd Qu.:12.586   3rd Qu.:127.2   3rd Qu.:1496   3rd Qu.: 52.25
 Max.   :20.703   Max.   :911.0   Max.   :2747   Max.   :175.00
 NA's   :10       NA's   :5       NA's   :5      NA's   :5
       UC              FFA         Glucose      LDL_plus_VLDL    MCP_1_phys
 Min.   : 153.0   Min.   : 45   Min.   :227.0   Min.   : 359   Min.   : 34.99
 1st Qu.: 390.0   1st Qu.: 91   1st Qu.:369.8   1st Qu.:1019   1st Qu.:105.52
 Median : 480.5   Median :108   Median :432.0   Median :1252   Median :136.18
 Mean   : 485.3   Mean   :115   Mean   :445.5   Mean   :1258   Mean   :153.02
 3rd Qu.: 565.0   3rd Qu.:136   3rd Qu.:511.0   3rd Qu.:1451   3rd Qu.:177.22
 Max.   :1289.0   Max.   :227   Max.   :866.0   Max.   :2693   Max.   :823.05
 NA's   :5        NA's   :5     NA's   :5       NA's   :5      NA's   :17
  Insulin_ug_l     Glucose_Insulin    Leptin_pg_ml     Adiponectin
 Min.   :  264.0   Min.   :0.01401   Min.   :  1160   Min.   : 0.803
 1st Qu.:  822.5   1st Qu.:0.08794   1st Qu.: 47754   1st Qu.: 4.799
 Median : 1895.0   Median :0.23072   Median : 74982   Median : 7.287
 Mean   : 4124.0   Mean   :0.33111   Mean   : 84814   Mean   : 8.410
 3rd Qu.: 5156.5   3rd Qu.:0.51633   3rd Qu.:111745   3rd Qu.:10.373
 Max.   :29699.0   Max.   :1.80682   Max.   :383322   Max.   :30.820
 NA's   :30        NA's   :31        NA's   :34       NA's   :41
 Aortic lesions         Note           Aneurysm       Aortic_cal_M
 Min.   :     63   Min.   :0.6000   Min.   :  0.00   Min.   : 0.00
 1st Qu.: 126750   1st Qu.:0.7500   1st Qu.:  0.00   1st Qu.: 0.00
 Median : 199750   Median :0.7500   Median : 14.50   Median : 2.00
 Mean   : 220089   Mean   :0.7417   Mean   : 19.39   Mean   : 3.85
 3rd Qu.: 277250   3rd Qu.:0.7500   3rd Qu.: 28.00   3rd Qu.: 6.00
 Max.   :3230750   Max.   :0.8500   Max.   :143.00   Max.   :35.00
 NA's   :8         NA's   :355      NA's   :15       NA's   :7
  Aortic_cal_L    CoronaryArtery_Cal Myocardial_cal    BMD_all_limbs
 Min.   : 0.000   Min.   : 0.0000    Min.   :   0.00   Min.   :0.04840
 1st Qu.: 0.000   1st Qu.: 0.0000    1st Qu.:   0.00   1st Qu.:0.05430
 Median : 0.000   Median : 0.0000    Median :   0.00   Median :0.05650
 Mean   : 1.903   Mean   : 0.7457    Mean   :  19.35   Mean   :0.05646
 3rd Qu.: 0.000   3rd Qu.: 0.0000    3rd Qu.:   2.00   3rd Qu.:0.05880
 Max.   :56.000   Max.   :47.0000    Max.   :3134.00   Max.   :0.06640
 NA's   :31       NA's   :70         NA's   :68        NA's   :142
 BMD_femurs_only
```

```
 Min.   :0.06380  
 1st Qu.:0.07590  
 Median :0.07975  
 Mean   :0.07974  
 3rd Qu.:0.08355  
 Max.   :0.09440  
 NA's   :142      
```

```r
## Check remove NAs for the entire data
liver_df_na <- na.omit(liver_df)

## Check remove NAs only in selected columns
liver_df_new <- drop_na(liver_df, c(weight_g, Trigly, Total_Chol, HDL_Chol,
                                    Glucose, Aneurysm))
```

2). Check the data further with skimr

```r
## Select column variables: Mice, Mouse_ID, Strain, sex, weight_g, Trigly,
## Total_Chol, HDL_Chol, Glucose, Aneurysm

liver_df_new <- liver_df_new |>
  select(c(Mice, Mouse_ID, Strain, sex, weight_g, Trigly, Total_Chol,
           HDL_Chol, Glucose, Aneurysm))

head(liver_df_new)
```

```
# A tibble: 6 x 10
  Mice   Mouse_ID Strain          sex weight_g Trigly Total_Chol HDL_Chol Glucose
  <chr>  <chr>    <chr>         <dbl>    <dbl>  <dbl>      <dbl>    <dbl>   <dbl>
1 F2_290 306-4    BxH ApoE-/-~      2     36.9     53       1167       50     437
2 F2_291 307-1    BxH ApoE-/-~      2     48.5     61       1230       32     572
3 F2_292 307-2    BxH ApoE-/-~      1     45.7     41       1285       81     497
4 F2_293 307-3    BxH ApoE-/-~      1     50.3    271       1299       64     553
5 F2_294 307-4    BxH ApoE-/-~      1     44.8    114       1410       50     535
6 F2_295 308-1    BxH ApoE-/-~      1     39.2     72       1533       18     382
# i 1 more variable: Aneurysm <dbl>
```

```r
## Check the data
## Skimr package with the skim() function provides more descriptive information
## about the data compared to summary(). You can see number of colums and rows;
```

```r
## number of columns that are character and numeric; missing rate per columns; etc

skim(liver_df_new)
```

Table 1: Data summary

Name	liver_df_new
Number of rows	337
Number of columns	10
Column type frequency:	
character	3
numeric	7
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Mice	0	1	4	7	0	337	0
Mouse_ID	0	1	5	6	0	337	0
Strain	0	1	11	15	0	3	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
sex	0	1	1.47	0.50	1.0	1.0	1.0	2.0	2.0	
weight_g	0	1	43.44	7.70	22.2	38.5	43.9	49.1	64.4	
Trigly	0	1	117.03	120.69	2.0	56.0	88.0	128.0	911.0	
Total_Chol	0	1	1311.02	339.62	380.0	1065.0	1299.0	1502.0	2747.0	
HDL_Chol	0	1	44.17	18.73	10.0	32.0	42.0	53.0	175.0	
Glucose	0	1	449.55	106.52	227.0	373.0	436.0	515.0	866.0	
Aneurysm	0	1	19.50	22.30	0.0	0.0	14.0	30.0	143.0	

3). Data filtering

```r
## See the first few rows of the filtered data for NAs
liver_df_new
```

```
# A tibble: 337 x 10
   Mice   Mouse_ID Strain        sex weight_g Trigly Total_Chol HDL_Chol Glucose
   <chr>  <chr>    <chr>       <dbl>    <dbl>  <dbl>      <dbl>    <dbl>   <dbl>
 1 F2_290 306-4    BxH ApoE-/~     2     36.9     53       1167       50     437
 2 F2_291 307-1    BxH ApoE-/~     2     48.5     61       1230       32     572
 3 F2_292 307-2    BxH ApoE-/~     1     45.7     41       1285       81     497
 4 F2_293 307-3    BxH ApoE-/~     1     50.3    271       1299       64     553
 5 F2_294 307-4    BxH ApoE-/~     1     44.8    114       1410       50     535
 6 F2_295 308-1    BxH ApoE-/~     1     39.2     72       1533       18     382
 7 F2_296 308-2    BxH ApoE-/~     2     38.8    120       1386       29     569
 8 F2_297 308-3    BxH ApoE-/~     2     31.2     15        699       10     376
 9 F2_298 308-4    BxH ApoE-/~     2     33.4     65       1334       19     568
10 F2_299 338-1    BxH ApoE-/~     2     40       83       1114       33     464
# i 327 more rows
# i 1 more variable: Aneurysm <dbl>
```

```r
## Filter by sex and create a data frame for each sex
liver_f <- liver_df_new |>
  filter(sex==1)

liver_m <- liver_df_new |>
  filter(sex==2)

## Check the first six rows for each sex
liver_df_new |>
  filter(sex==1) |>
  head()
```

```
# A tibble: 6 x 10
  Mice   Mouse_ID Strain        sex weight_g Trigly Total_Chol HDL_Chol Glucose
  <chr>  <chr>    <chr>       <dbl>    <dbl>  <dbl>      <dbl>    <dbl>   <dbl>
1 F2_292 307-2    BxH ApoE-/--~     1     45.7     41       1285       81     497
2 F2_293 307-3    BxH ApoE-/--~     1     50.3    271       1299       64     553
3 F2_294 307-4    BxH ApoE-/--~     1     44.8    114       1410       50     535
4 F2_295 308-1    BxH ApoE-/--~     1     39.2     72       1533       18     382
```

```
5 F2_308 374-3    BxH ApoE-/-~      1     40.5       7        1284        28      575
6 F2_313 341-1    BxH ApoE-/--      1     46.2       5        1863        54      460
# i 1 more variable: Aneurysm <dbl>
```

```r
liver_df_new |>
  filter(sex==2) |>
  head()
```

```
# A tibble: 6 x 10
  Mice   Mouse_ID Strain         sex weight_g Trigly Total_Chol HDL_Chol Glucose
  <chr>  <chr>    <chr>        <dbl>    <dbl>  <dbl>      <dbl>    <dbl>   <dbl>
1 F2_290 306-4    BxH ApoE-/--     2     36.9     53       1167       50     437
2 F2_291 307-1    BxH ApoE-/~      2     48.5     61       1230       32     572
3 F2_296 308-2    BxH ApoE-/--     2     38.8    120       1386       29     569
4 F2_297 308-3    BxH ApoE-/~      2     31.2     15        699       10     376
5 F2_298 308-4    BxH ApoE-/~      2     33.4     65       1334       19     568
6 F2_299 338-1    BxH ApoE-/--     2     40       83       1114       33     464
# i 1 more variable: Aneurysm <dbl>
```

4). Arrange the data by a specific column

```r
## Arrange by weight_g in descending order
## Note you can arrange by more than 1 column by comma separating the columns.
## E.g arrange(desc(weight_g, Trigly, Total_Chol)). However, the first selected
## column takes precedence, then the second, and third.

liver_df_new |>
  arrange(desc(weight_g)) |>
  head()
```

```
# A tibble: 6 x 10
  Mice   Mouse_ID Strain         sex weight_g Trigly Total_Chol HDL_Chol Glucose
  <chr>  <chr>    <chr>        <dbl>    <dbl>  <dbl>      <dbl>    <dbl>   <dbl>
1 F2_146 259-1    BxH ApoE-/~      1     64.4    438       1671       48     703
2 F2_173 268-1    BxH ApoE-/--     1     60.7    107       1732       35     437
3 F2_35  154-2    BxH ApoE-/--     1     59.4     79       2747       54     780
4 F2_5   134-2    BxH ApoE-/~      1     59.2    261       1563       55     594
5 F2_73  182-1    BxH ApoE-/--     1     59.2     82       1714       41     592
6 F2_10  136-1    BxH ApoE-/~      1     58.4    134       1567       45     564
# i 1 more variable: Aneurysm <dbl>
```

5). Data summarizing

```
## Summarize weight_g and Glucose level by strain

liver_df_new |>
  group_by(Strain) |>
  summarise(n=n(),
            avg_weight=mean(weight_g),
            avg_gluc=mean(Glucose))
```

```
# A tibble: 3 x 4
  Strain                n avg_weight avg_gluc
  <chr>             <int>      <dbl>    <dbl>
1 BxH ApoE-/-, F1       8       46.0     404.
2 BxH ApoE-/-, F2     322       43.4     449.
3 C3H ApoE-/-           7       43.0     537
```

```
## Summarise the mean of all column variables

liver_df_new |>
  group_by(Strain) |>
  summarise(across(where(is.numeric),
                   mean))
```

```
# A tibble: 3 x 8
  Strain              sex weight_g Trigly Total_Chol HDL_Chol Glucose Aneurysm
  <chr>             <dbl>    <dbl>  <dbl>      <dbl>    <dbl>   <dbl>    <dbl>
1 BxH ApoE-/-, F1    1.5      46.0   46.5      1226.     43.2    404.    12.5
2 BxH ApoE-/-, F2    1.48     43.4  109.       1315.     43.9    449.    19.9
3 C3H ApoE-/-        1        43.0  565.       1242.     59.7    537      7.29
```

6). Data pivoting

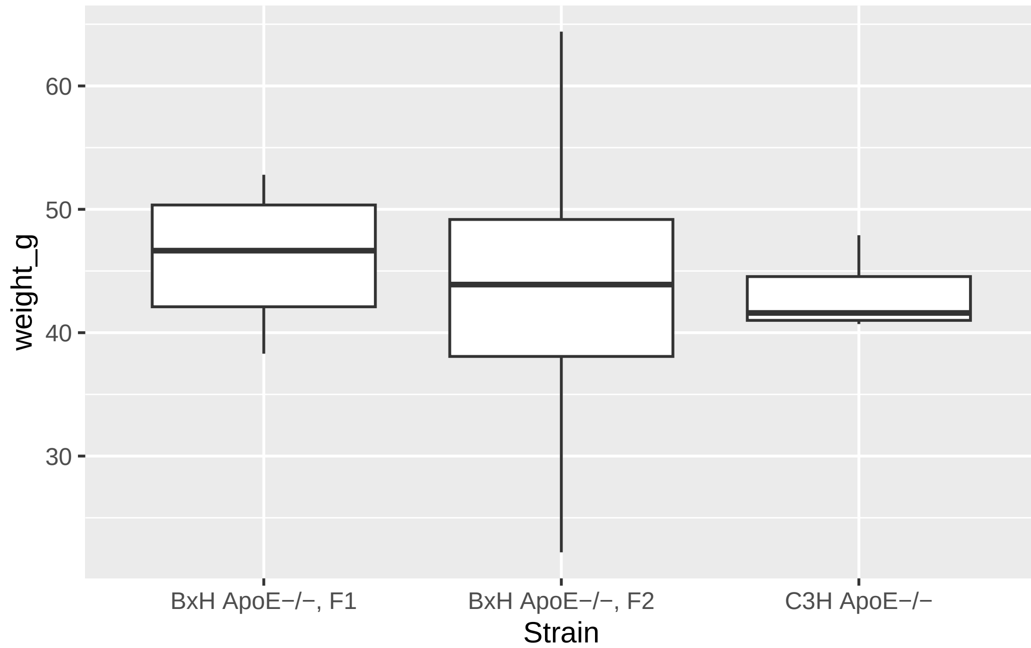
```
## Pivoting

## With pivoting, we're transforming by either lengthening the rows (pivot_longer)
## and reducing the columns or widening the columns (pivot_wider) and reducing
## the rows
```

```r
liver_lng <- liver_df_new |>
  pivot_longer(cols = weight_g:Aneurysm,
               names_to = "variables",
               values_to = "value")
head(liver_lng)
```

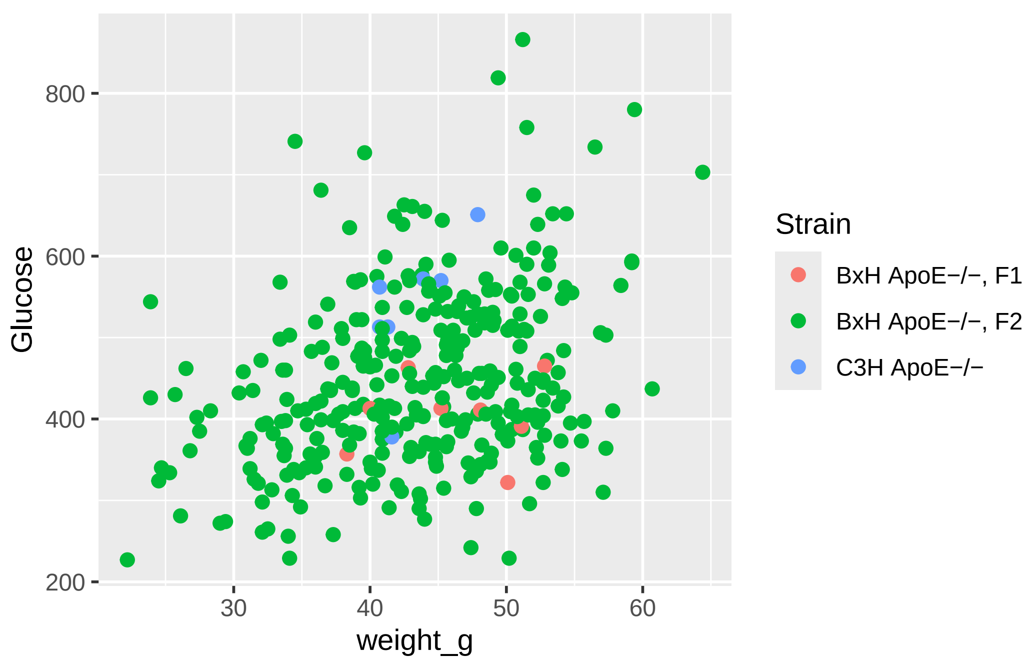
```
# A tibble: 6 x 6
  Mice   Mouse_ID Strain              sex variables  value
  <chr>  <chr>    <chr>             <dbl> <chr>      <dbl>
1 F2_290 306-4    BxH ApoE-/-, F2       2 weight_g    36.9
2 F2_290 306-4    BxH ApoE-/-, F2       2 Trigly      53
3 F2_290 306-4    BxH ApoE-/-, F2       2 Total_Chol 1167
4 F2_290 306-4    BxH ApoE-/-, F2       2 HDL_Chol    50
5 F2_290 306-4    BxH ApoE-/-, F2       2 Glucose    437
6 F2_290 306-4    BxH ApoE-/-, F2       2 Aneurysm    16
```

7). Data Visualization

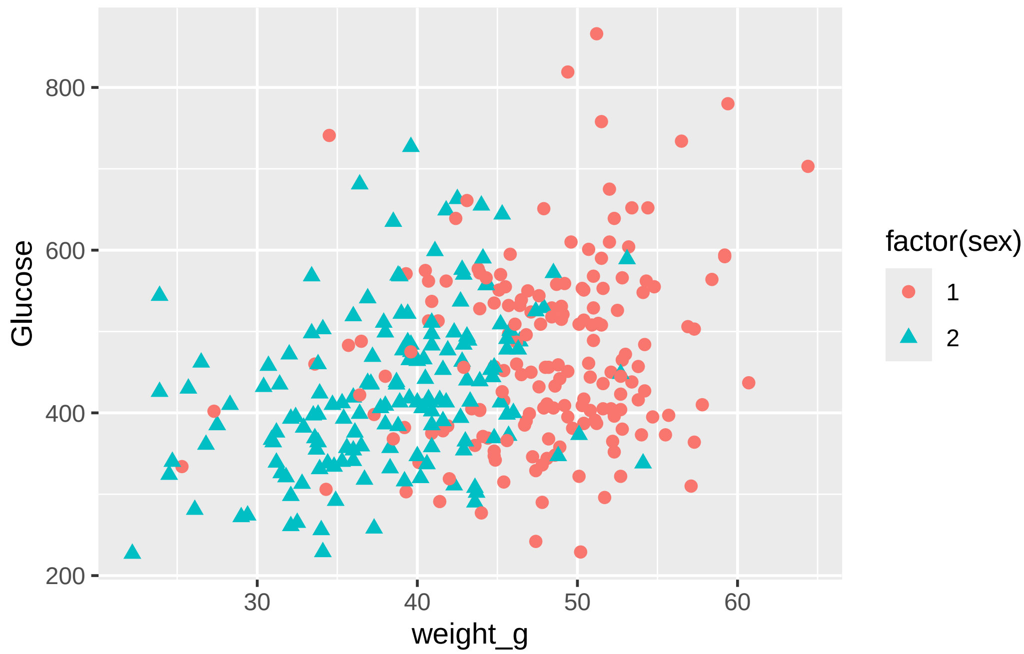
```r
#######################################
## Plot with ggplot2 package
######################################

ggplot(liver_df_new, aes(Strain, weight_g))+
  geom_boxplot()
```



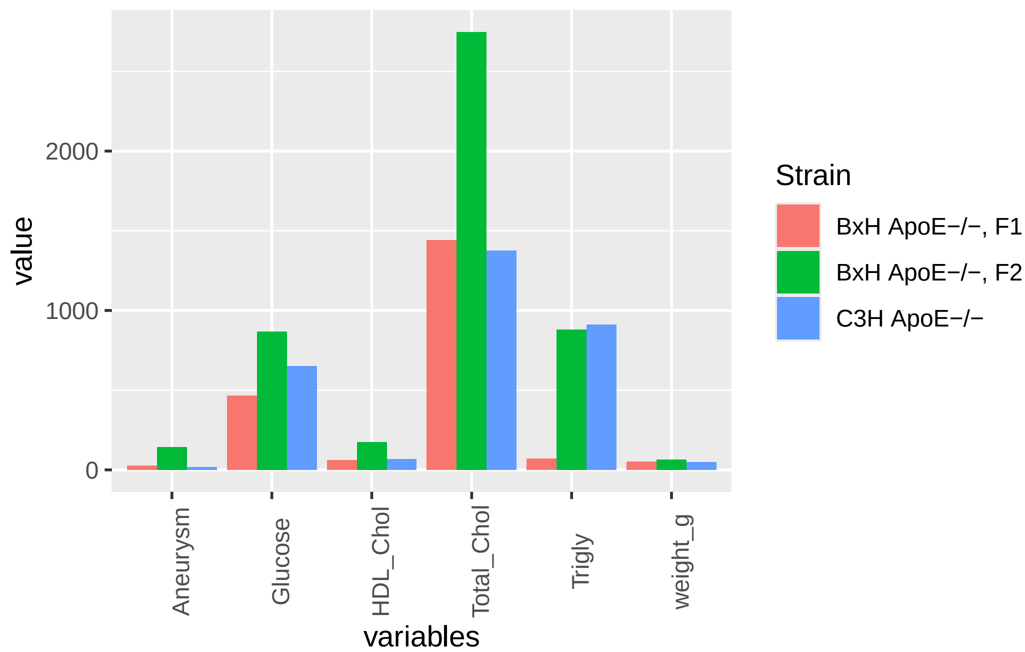

```r
ggplot(liver_df_new, aes(weight_g, Glucose, color = Strain))+
  geom_point(size=2)
```



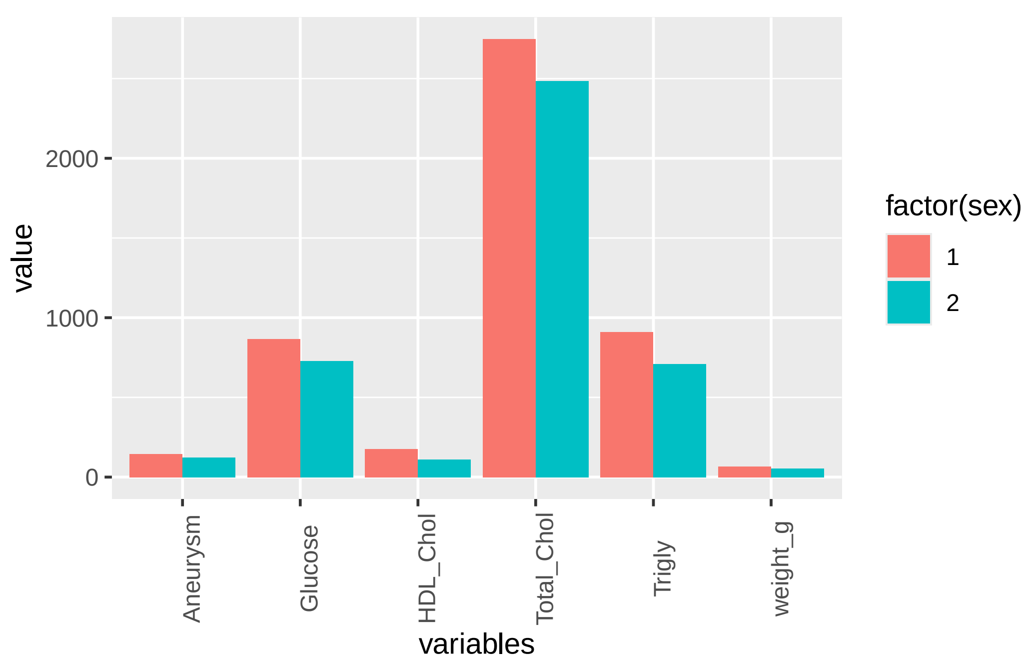
```r
ggplot(liver_df_new, aes(weight_g, Glucose, shape = factor(sex)))+
  geom_point(size=2, aes(color = factor(sex)))
```




```r
ggplot(liver_lng, aes(variables, value, fill =Strain ))+
  geom_col(position="dodge")+
  theme(axis.text.x = element_text(angle = 90))
```

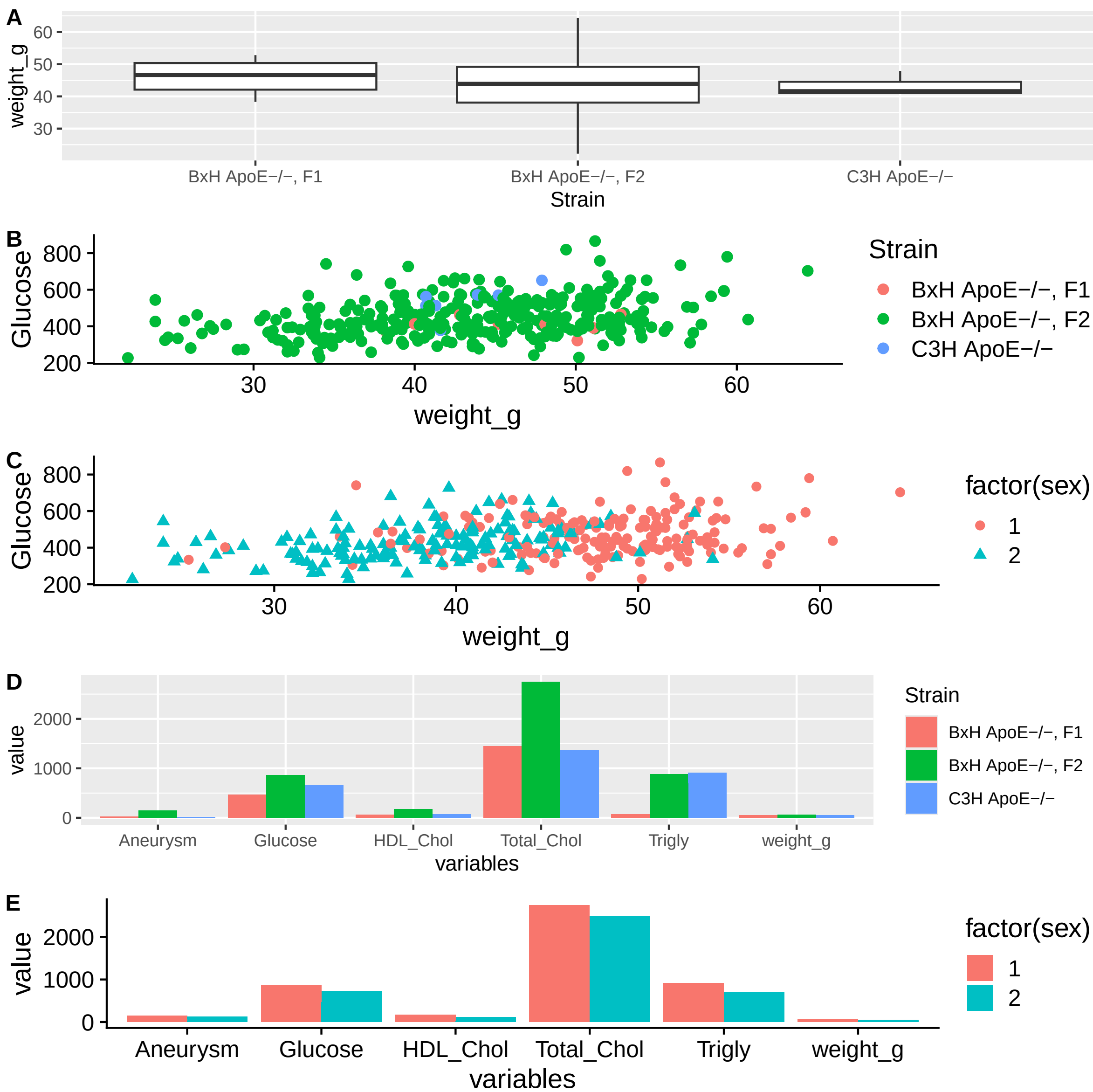



```
ggplot(liver_lng, aes(variables, value, fill = factor(sex)))+
  geom_col(position="dodge")+
  theme(axis.text.x = element_text(angle = 90))
```



8). Data visualization: Publication-ready

```r
########################################
## Plot with ggplot2 and cowplot package
#######################################

g1 <- ggplot(liver_df_new, aes(Strain, weight_g))+
  geom_boxplot()

g2 <- ggplot(liver_df_new, aes(weight_g, Glucose, color = Strain))+
  geom_point(size=2)+
  theme_cowplot()

g3 <- ggplot(liver_df_new, aes(weight_g, Glucose, shape = factor(sex)))+
  geom_point(aes(color = factor(sex)),
             size=2)+
  theme_cowplot()

g4 <- ggplot(liver_lng, aes(variables, value, fill =Strain ))+
  geom_col(position="dodge")

g5 <- ggplot(liver_lng, aes(variables, value, fill =factor(sex) ))+
  geom_col(position="dodge")+
  theme_cowplot()

plot_grid(g1, g2, g3, g4, g5, labels = c("A", "B", "C", "D", "E"), ncol = 1,
          label_size = 12)
```



9). Data tabularization

```
liver_tbl <- liver_df_new |>
             group_by(Strain) |>
             select(-4) |>
             summarise(across(where(is.numeric),
                              mean))
```

```r
liver_ft <- flextable(liver_tbl)

set_table_properties(liver_ft, layout = "autofit")
```

Strain	weight_g	Trigly	Total_Chol	HDL_Chol	Glucose	Aneurysm
BxH ApoE-/-, F1	46.05000	46.5000	1,225.750	43.25000	404.3750	12.500000
BxH ApoE-/-, F2	43.38065	109.0435	1,314.630	43.85404	448.7764	19.934783
C3H ApoE-/-	43.04286	565.1429	1,242.286	59.71429	537.0000	7.285714

```r
## Save to Word file. Note you can also save to image, powerpoint, etc
#save_as_docx(liver_ft, path = "")
```

10). Data visualization: Publication-ready

```r
#########################################
## Data
########################################

## Load the mammalians sleep dataset
data("msleep")
sleep_df <- msleep |>
  na.omit() |>
  select(-2)

## Remove the "sleep_" from column names
names(sleep_df) <- str_replace_all(names(sleep_df),"sleep_", "")

#######################################
## Plot with ggplot2 and cowplot package
######################################

## Read different data information into an object for plotting

S1 <- sleep_df |>
  ggplot(aes(total, fill = vore))+
    geom_density(alpha=0.5)+
    theme_cowplot(12)
```

```r
S2 <- sleep_df |>
  mutate(sleep_nrem = (1-(rem/total))*total)|>
  ggplot(aes(rem, sleep_nrem, color = vore))+
    geom_point(size=2)+
    theme_cowplot(12)

S3 <- sleep_df |>
  ggplot(aes(awake, total, color = vore))+
    geom_point(size=2)+
    theme_cowplot(12)

S4 <- sleep_df |>
  ggplot(aes(vore,bodywt))+
    geom_boxplot()+
    theme_cowplot(12)

plot_grid(S1, S2, S3, S4, labels = c("A", "B", "C", "D"))
```